DO YOU KNOW?

1 cm = 1/100 meter
1 mm = 1/1000 meter =1/10 cm
1 µm = 1/1000,000 meter = 1/10,000 cm
1 nm = 1/1,000,000,000 meter = 1/10,000,000 cm
1 Å = 1/10,000,000,000 meter =1/100,000,000 cm

or

$1\ m = 10^2 cm = 10^3 mm = 10^6 \mu m = 10^9 nm = 10^{10} Å$

m = meter cm = centimetre mm = millimeter µm = micrometer
nm = nanometer Å=Angstrom

vary from round (**cocci**) to rectangular (**rod**). In virus, shape of the envelope varies from round to hexagonal or '**T**' shaped. In fungi, globular to elongated cylindrical cells and the spores of fungi vary greatly in shape. In plants and animals cells vary in shape according to cell types such as parenchyma, mesophyll, palisade, tracheid, fiber, epithelium and others (Figure 6.6).

Size:

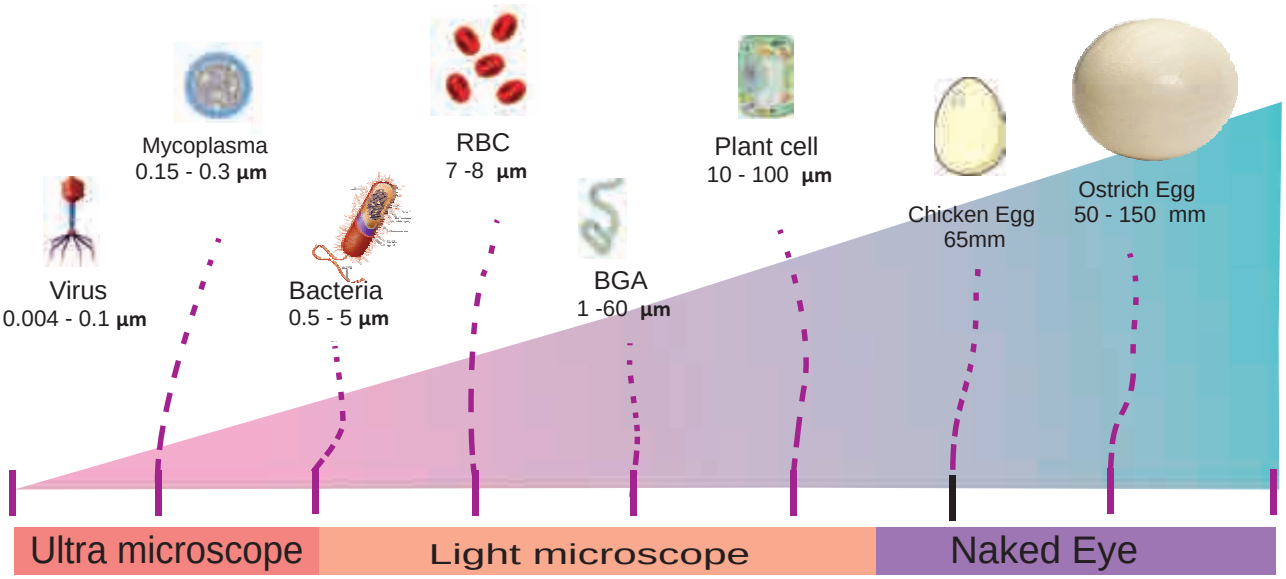


Figure 6.6: Cell size variation of few organisms

6.4. Types of cells

On the basis of the cellular organization and the nuclear characteristics, the cell can be divided into

- Prokaryotes
- Mesokaryotes and
- Eukaryotes

6.4.1 Prokaryotes

Those organisms with primitive nucleus are called as **prokaryotes** (*pro* – primitive; *karyon* – nucleus). The DNA lies in the 'nucleoid' which is not bound by the nuclear membrane and therefore it is not a true nucleus and is also a primitive type